

Insight AI – An Intelligent AI-Powered Data Analytics Web Application

SYNOPSIS

Submitted By

Aman Singh(22BTCSE1010)

Hari Om(22BTCSE1019)

Dipesh Singh(22BTCSE0330)

Harsh Thapa(22BTDATA1001)

Under the Guidance of

Department of Computer Science & Engineering

School of Engineering & Computing (SoEC)



Dev Bhoomi Uttarakhand University

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ABSTRACT

In recent years, the rapid growth of data has increased the demand for efficient and accessible data analytics solutions across various domains. However, most traditional data analytics tools require significant technical expertise in programming and statistical analysis, which limits their usability for non-technical users.

In this paper, we propose “**Insight AI – An Intelligent AI-Powered Data Analytics Web Application**”, a novel system designed to simplify the process of data analysis through automation and natural language interaction. The system enables users to upload datasets and perform data analysis using intuitive natural language queries without requiring any coding knowledge.

The proposed application integrates Artificial Intelligence and Large Language Models (LLMs) to interpret user queries and dynamically generate analytical outputs and visualizations. It also incorporates automated data cleaning, interactive dashboards, and report generation functionalities. The system is implemented using modern technologies such as Python, Streamlit, Supabase, and Plotly.

Experimental results demonstrate that the system provides accurate, efficient, and real-time insights while significantly reducing manual effort. The proposed solution enhances user accessibility and simplifies complex analytical processes, making data analytics more inclusive.

Overall, Insight AI represents a step towards the automation and democratization of data analytics by combining intelligent systems with user-friendly interfaces.

Keywords: Artificial Intelligence, Data Analytics, Natural Language Processing (NLP), Large Language Models (LLMs), Data Visualization, Machine Learning

1. INTRODUCTION

In the modern digital era, data has become a critical asset for organizations, researchers, and individuals. The rapid growth of data generated from various sources such as business transactions, social media, and online platforms has created a need for efficient data analytics systems. Extracting meaningful insights from raw data is essential for informed decision-making and strategic planning.

However, traditional data analytics tools often require technical expertise in programming languages such as Python or R, along with knowledge of statistical methods. This creates a barrier for non-technical users who need to analyze data but lack coding skills. Additionally, the process of data cleaning, analysis, and visualization is often time-consuming and requires manual effort.

With the advancement of Artificial Intelligence (AI), particularly Natural Language Processing (NLP) and Large Language Models (LLMs), new possibilities have emerged for simplifying data analytics. These technologies enable systems to understand human language and convert user queries into meaningful analytical operations.

In this context, we propose **Insight AI – An Intelligent AI-Powered Data Analytics Web Application**, which aims to automate the entire data analytics pipeline. The system allows users to upload datasets and interact with data using natural language queries. It performs automated data cleaning, generates interactive visualizations, and provides a dashboard for managing insights.

The proposed system focuses on improving accessibility, reducing complexity, and enhancing user experience. By integrating AI with data analytics, Insight AI provides a scalable and efficient solution for modern data analysis challenges

2. LITERATURE REVIEW

Data analytics has been widely explored through various tools and technologies over the years. Traditional tools such as Microsoft Excel, Tableau, and Power BI have been extensively used for data processing and visualization. These tools provide user-friendly interfaces but often require manual effort in data cleaning and lack automation capabilities.

Python-based data analytics systems have gained popularity due to their flexibility and powerful libraries such as Pandas, NumPy, and Matplotlib. These systems enable advanced data analysis and visualization; however, they require programming knowledge and technical expertise, making them less accessible to non-technical users.

With the advancement of Artificial Intelligence, especially Natural Language Processing (NLP) and Large Language Models (LLMs), new approaches to data analytics have emerged. AI-based systems can interpret user queries and automate the analysis process, reducing manual effort and improving efficiency.

Recent research has focused on developing intelligent systems that integrate AI with data analytics. However, many of these systems either lack user-friendly interfaces or do not provide a complete end-to-end solution that includes data cleaning, analysis, visualization, and reporting.

The review of existing work highlights a clear gap in the development of a fully automated and accessible data analytics platform. This gap motivates the design of the proposed system, Insight AI, which aims to combine automation, intelligence, and ease of use into a single integrated solution.

3. METHODOLOGY

The proposed system, Insight AI, is designed as an intelligent web-based application that automates the data analytics process using Artificial Intelligence and Natural Language Processing. The system follows a structured architecture consisting of multiple components that work together to provide seamless data analysis and visualization.

The system operates in four major phases: (1) User Authentication, (2) Data Ingestion and Profiling, (3) Data Cleaning and AI-Based Analytics, and (4) Visualization and Dashboard Management. These phases ensure a structured and automated workflow.

In Phase 1, the user logs into the system using secure authentication provided by Supabase. This ensures data privacy and personalized access for each user.

In Phase 2, the user uploads a dataset in CSV or Excel format. The system reads the data using Pandas and performs initial data profiling, including summary statistics, data types, and preview of the dataset.

In Phase 3, the system performs automated data cleaning operations such as handling missing values and removing duplicates. After preprocessing, the user interacts with the system using natural language queries. The AI engine, powered by Groq API, Google Gemini API, and LangChain, interprets the query and generates executable Python code for data analysis.

In Phase 4, the processed data is converted into interactive visualizations using Plotly. The generated charts are displayed to the user and can be saved to a dashboard. The results are stored in the Supabase database, and users can export the analysis in the form of reports.

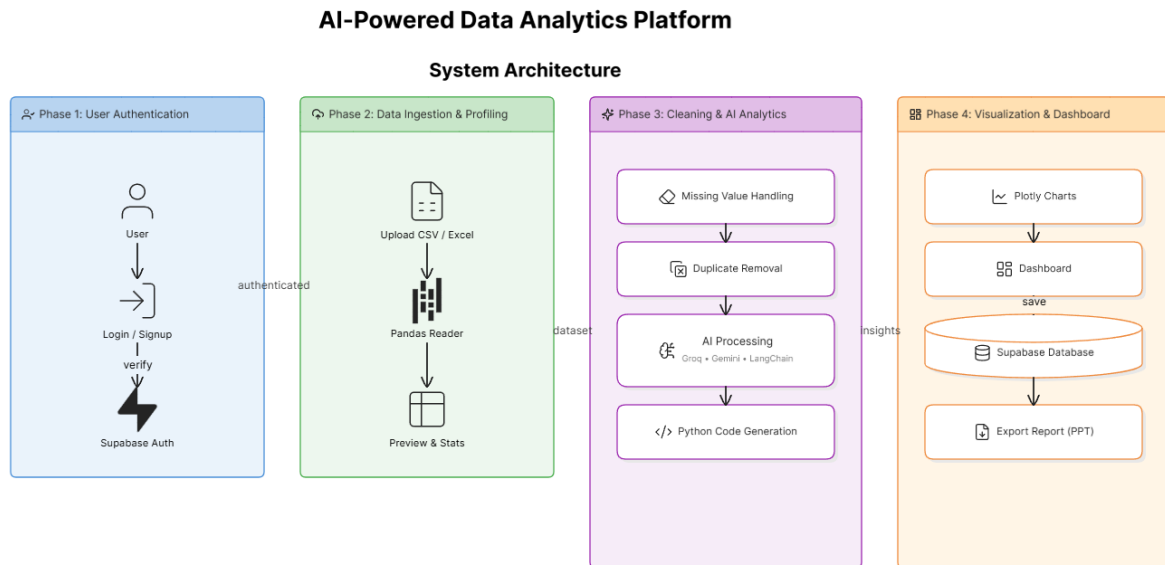


Figure 1: System Architecture of Insight AI

The above figure illustrates the overall architecture of the system, showing the interaction between the user interface, AI engine, data processing modules, and database.

The system workflow follows a sequential process starting from user login to final output generation. Each stage of the workflow is designed to minimize manual effort and enhance user experience.

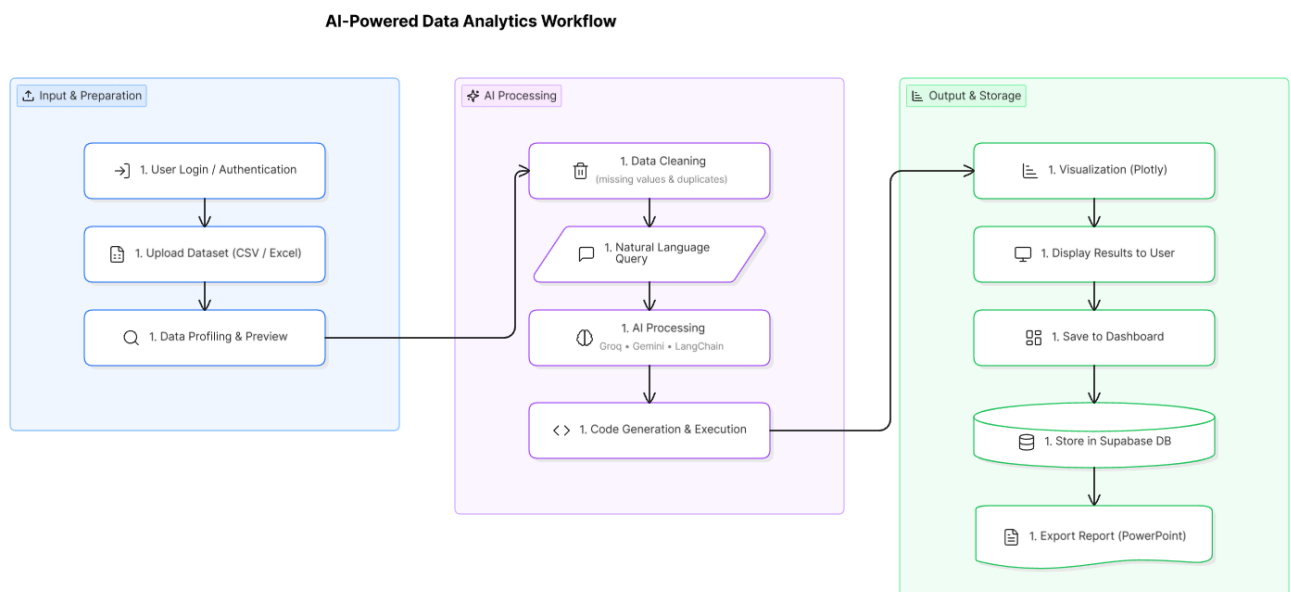


Figure 2: Workflow of Insight AI System

The workflow diagram represents the step-by-step process of the system, including data ingestion, preprocessing, AI-based analysis, visualization, and dashboard storage.

4. IMPLEMENTATION

The Insight AI system is implemented using a combination of modern technologies to ensure efficiency, scalability, and ease of use. The application is developed using Python as the core programming language due to its extensive support for data analytics and machine learning.

The frontend of the system is built using Streamlit, which provides an interactive and user-friendly interface. It allows users to upload datasets, interact with the system using natural language queries, and visualize results in real time.

The backend is powered by Supabase, which handles user authentication, database management, and secure storage of user data. It ensures data privacy through row-level security and session management.

The system integrates Artificial Intelligence through Groq API, Google Gemini API, and LangChain. These technologies enable the system to process natural language queries and generate executable Python code dynamically.

Data processing is performed using libraries such as Pandas and NumPy, while visualization is achieved using Plotly, Matplotlib, and Seaborn. The system also supports report generation using python-pptx.

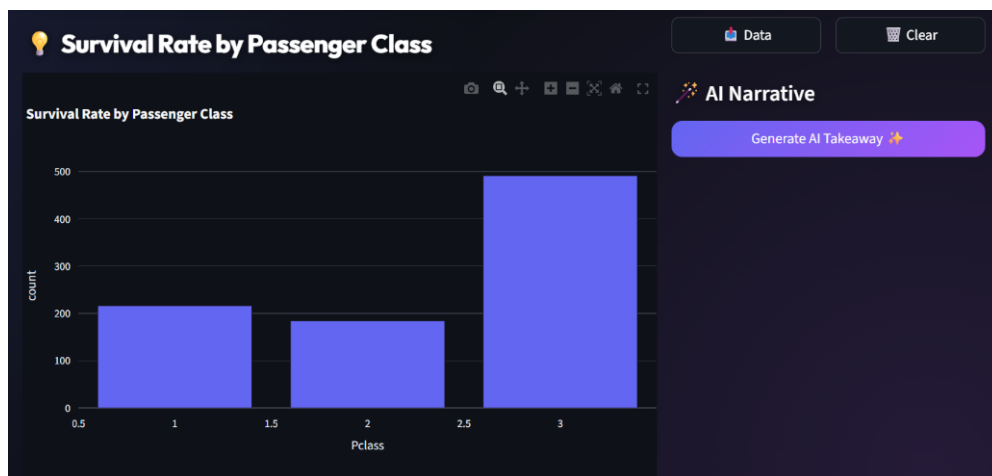
Overall, the implementation ensures seamless integration of multiple components to provide a complete and automated data analytics solution.

5. RESULTS AND DISCUSSION

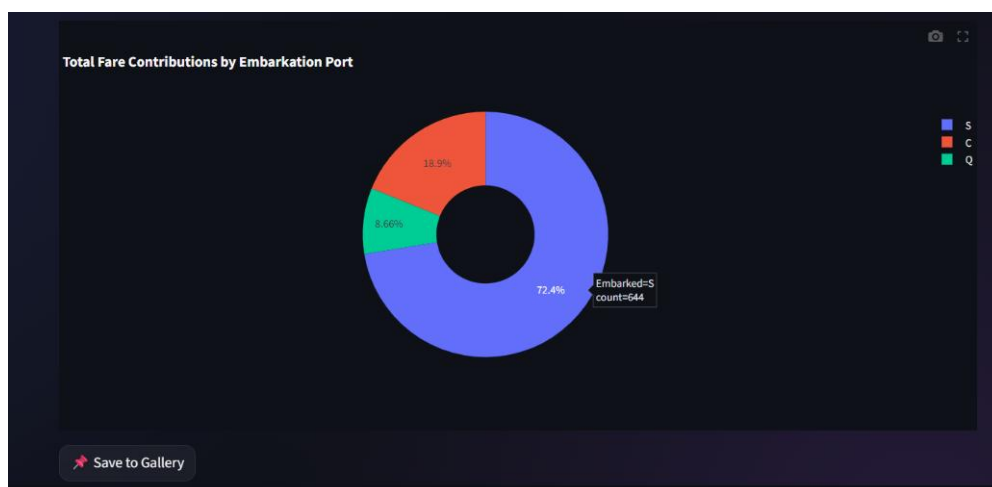
The proposed system was tested using multiple datasets from different domains such as sales, student performance, and finance. The results demonstrate that the system is capable of processing datasets efficiently and generating meaningful insights.

The system successfully performed data cleaning operations, including handling missing values and removing duplicates. It also generated accurate and interactive visualizations based on user queries.

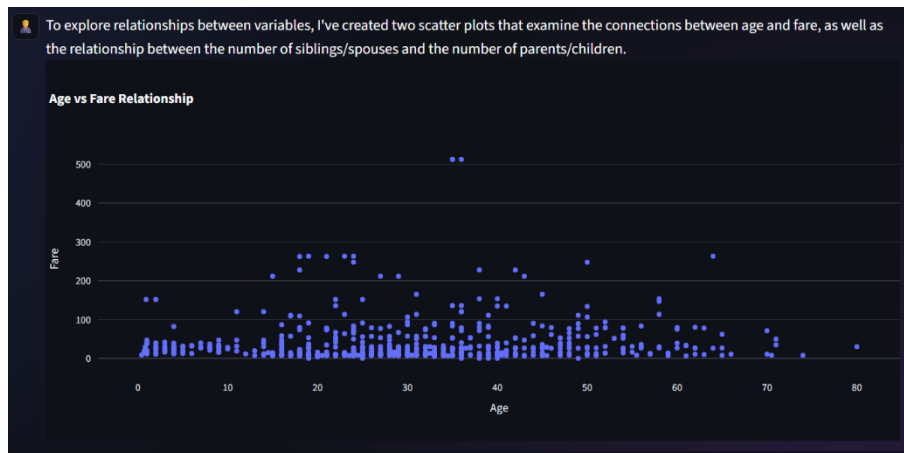
The AI module was able to interpret natural language queries and convert them into executable operations, providing real-time analytical results. The response time of the system was found to be satisfactory for most queries.



Bar graph



Pie chart



Scatter plot

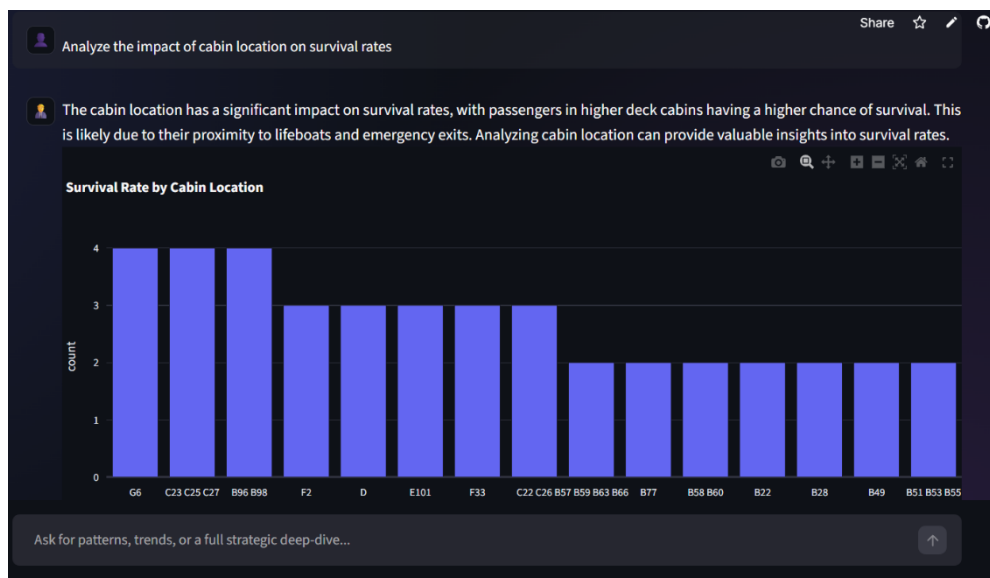


Figure 3: Visualization Generated by Insight AI

The above figure shows an example of the visualization generated by the system. The charts are interactive and help users understand data patterns effectively.

Overall, the system reduces manual effort and improves the efficiency of data analysis, making it suitable for non-technical users

6. CONCLUSION

In this paper, we presented Insight AI, an intelligent AI-powered data analytics web application designed to simplify the process of data analysis. The system successfully integrates artificial intelligence with data analytics to provide a no-code solution for users.

The proposed system enables users to upload datasets, perform data analysis using natural language queries, and generate interactive visualizations. It automates key processes such as data cleaning, analysis, and reporting, thereby reducing complexity and effort.

The results demonstrate that the system is efficient, accurate, and user-friendly. It enhances accessibility by allowing non-technical users to perform data analysis without programming knowledge.

Overall, Insight AI provides a scalable and effective solution for modern data analytics challenges and contributes to the advancement of AI-driven systems.

7. FUTURE SCOPE

The system can be further enhanced by integrating real-time data sources such as APIs and IoT devices. Advanced machine learning models can be incorporated to provide predictive analytics and forecasting capabilities.

The development of a mobile application can improve accessibility and usability. Additionally, collaboration features can be added to support multiple users working on shared datasets.

Further improvements can include enhanced visualization customization and offline functionality to reduce dependency on internet connectivity.

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